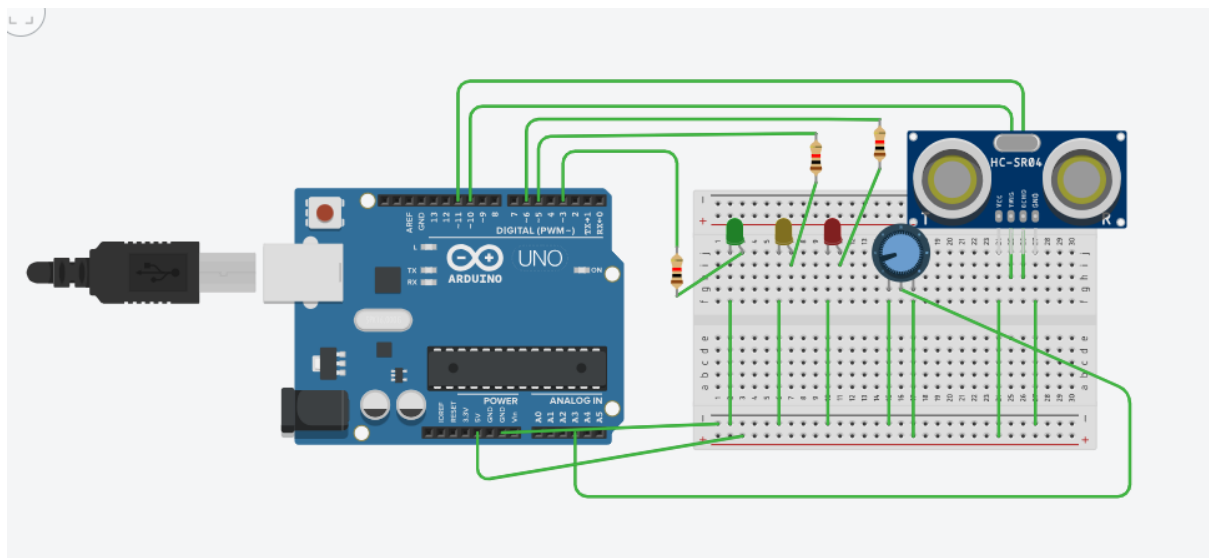


Robocon- Tinkercad Task-3.2

Circuit diagram and code explanation

Task- connect the ultrasonic sensor, potentiometer, and three LEDs to the Arduino and write a program that reads the distance from the ultrasonic sensor and the value of the potentiometer and controls the LEDs according to the given conditions

Circuit:



Explanation:

Components—Arduino UNO, breadboard, ultrasonic distance sensor, 3 LEDs, 3 resistors, potentiometer, jumper wires

Code logic—

- We take input value from potentiometer and store it in a variable
- We send a pulse through trig and observe the duration of echo in high then with that duration we calculate the distance at which the obstacle is

- We convert the 10 bit input value of potentiometer to 10 to 50 and name it warning distance as per the question and control the LEDs using if else condition
 - If the distance > warning distance then green led glows rest are off
 - For distance > warning distance/2 yellow led glows and rest LEDs are at rest
 - When distance < warning distance/2 red led glows remaining stay off
- We print the distance, warning distance, robot status for all condition using Serial.print